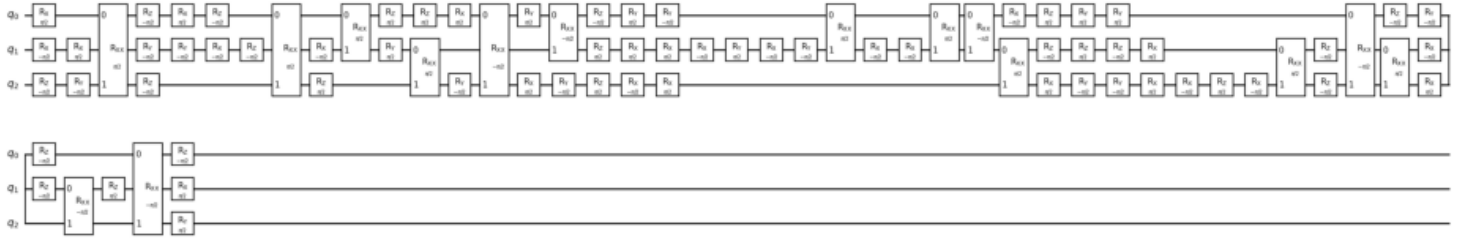
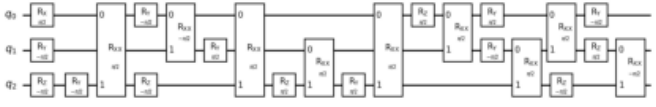
(a) input: random 3-qubit word over the ion alphabet, 80 gates



(b) original method (database radius 5): 24 gates



(c) proven optimum (exact Cayley table): 11 gates